

The effects of topology and task complexity on digital-speech-desktop interface design and evaluation

Wen-Te Chang
Chien-Hsiung Chen

Abstract — This study investigated user task performance in terms of three interface proposals (linear, hierarchical, and network type) and task complexities (single-layered, double-layered, and triple-layered). Forty-two participants (all male), aged 19–23 years, were recruited for the experiment. The results generated from this study reveal that the different degrees of task complexity interacted with the topological structures of the user interfaces. No difference was found among the three topological structures in the single-layered task. A linear structure resulted in better user performance than network or hierarchical structures in the double-layered task, while in the triple-layered task both network and hierarchical structures resulted in better user performance than a linear topological structure. This study provides an example of the application of topological structures in interface design and evaluation. From a practical perspective, the results imply the necessity of alternative or parallel control systems (linear and hybrid topologies) so that the user can shift between structures according to task complexity.

Keywords — Task complexity, user-interface design, topology, digital speech desktop.

DOI # 10.1889/JSID19.10.700

1 Introduction

Interface and display technology is developing rapidly and context dependently. Applications such as digital educative systems in classrooms, particularly digital speech desktops for classrooms, have been implemented and continuously updated in most Taiwanese educational institutions. The majority has implemented digital speech systems that include functions such as microphones, audio, projectors, and digital blackboards, and only a few conventional push-button interfaces remain in use. This new type of interface representation has been extremely effective, moving well beyond people's current experience.^{1–3} Layouts and topological structures have developed rapidly, yet uncertainty remains over how to determine the best design among all the design proposals available in terms of different interface topological structures. A problematic design may affect the educational effectiveness and quality of an interface,⁴ and consequently, addressing interface design issues prior to evaluation of a system is crucial.^{5–8}

1.1 Topological structure

The concept of interface usability relates to how easily and effectively a user can interact with an interface to accomplish his/her goals.^{7,9–11} A new interface will need pilot tests or user-experience observations to discover how users learn to use the interface, whether based on instruction, previous

experience (of conventional interfaces), or through trial and error. The interface can then be modified or approved accordingly. Many types of interface design structures have been developed, although most are linear, hierarchical, or network structures.^{12,13} A linear topology is essentially characterized by linear links; the user can be clearly guided within this rigid structure (*i.e.*, less disorientation), but she or he has to endure tedious forward tracking and backtracking. Similar to a linear topology, a hierarchical topology consists of several linear functions grouped by the parent–child structure. The grouped linear functions can be effectively layered; thus, less memory is needed than for the linear topological structure, but users may still get lost due to the divergent functional branches. In contrast, a network structure provides users with all the functions and layouts in the same user interface, enabling them to visualize everything at once. This may enable faster navigation but may also result in cognitive overload as users attempt to access a specific hypertext within the various types of information and find that they do not know where to go.

Previous studies have attempted to find the best topology by comparing the merits of the various topologies available, but results have been divided as to which topological structure is easiest to use and best enhances effectiveness.¹⁴ Other studies have suggested that as structural depth increases, so does the response time to locate the target item.^{15–17} It has also been suggested that task complexity is one of the major factors underlying the uncertainty of the interface design comparisons and evaluations.^{18,19}

Received 02-28-11; accepted 07-06-11.

W-T. Chang is with Ming Chuan University, Department of Product Design, 5 De-Ming Rd., Gui-Shan, Taoyuan, Taiwan, 333 ROC; telephone +886-2-2882-4564 x5259, e-mail: wtchang@mail.mcu.edu.tw.

C-H. Chen, National Taiwan University of Science and Technology, Taiwan.

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1.2 Task complexity

A multi-functional or multi-layered task that requires more functional links and operations is more complex than a single-function or single-layered task that requires only a single link or function. A previous study found that a multi-layered task requiring the cross-referencing of information from more than one function could be carried out using less search time and fewer function layers in a network topology than in a linear topology. By contrast, a single-layered task requiring information from only a single linking function could be carried out more efficiently in a linear topology, with less memory load required for unrelated information compared with the network topology.²⁰ Given the correlation between topological structure and task complexity, in the present study, we hypothesized that the interaction between task type and topological structure would affect the comparison of interface designs. More specifically, linear topological structures should better support users in carrying out less-complex tasks, whereas network topological structures should better facilitate performance of tasks with greater complexity. To extend the potential scope and application of interface design and usability research, we adopted and tested designs and proposals for a digital speech desk-top interface.

To summarize, this study investigated users' task performance in terms of three different interface structural proposals (namely, linear, hierarchical, and network) and task complexities (single-, double-, and triple-layered). These results have implications for interface design relating to task type and topological structure, and they can be applied to user-interface design and evaluation.

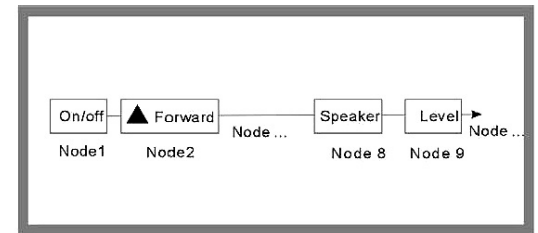
The following specific results were hypothesized:

- (1) Task complexity would affect the task performance of interface users.
- (2) There would be an interaction between topological structure and task complexity.

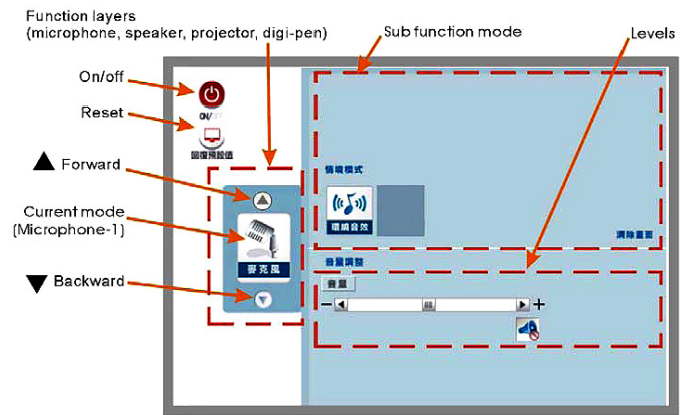
2 Method

2.1 Participants

As previous studies have argued that gender may affect judgment on navigational or wayfinding tasks, the present study invited only male participants to avoid any gender effect.²¹ Forty-two male participants were recruited for the experiment. All were college students from the Design School of Ming Chuan University, Taiwan, who spent an average of 5.09 hours ($SD = 2.34$) daily using a computer. Their ages ranged from 19 to 23 years old, with a mean age of 20.55 ($SD = 1.19$). All participants were paid approximately US\$6 per hour for their participation.



(b) Linear topological structure



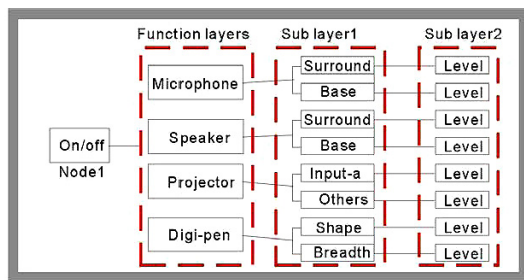
(a) Interface layout of linear design

FIGURE 1 — The interface layouts of the linear typological structure.

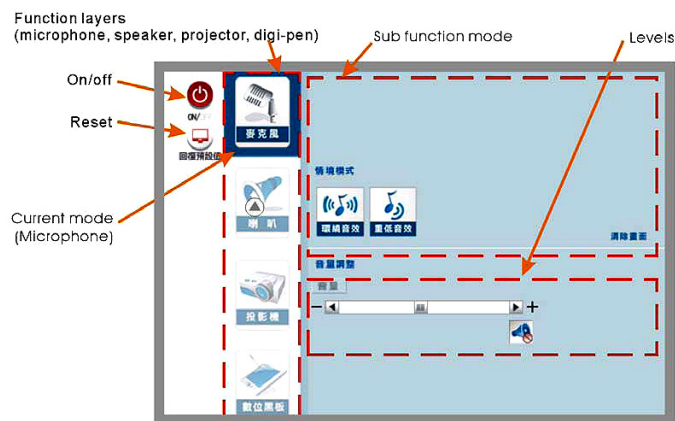
2.2 Stimulus materials and design

The designs for the digital-speech-desk-top user interfaces were implemented based on a functional investigation of the requirements of the teaching facility of Ming Chuan University, Taiwan. Five major user-interface functions were defined according to the results of the preliminary investigation: on/off switch, microphone, speaker, projector, and digi-pen modes. The number of functional nodes was 32.

The two independent variables were topological structure and task complexity. The levels of each variable are described below. The topological structures of the interface were linear, hierarchical, and network. For the linear type of interface structure, four nodes were shown on each level, and only one function could be seen and operated at a time [Fig. 1(a)]. Users could shift to other function modes by clicking on the linear-linked back or forward buttons [Fig. 1(b)]. For the hierarchical type of interface structure, two levels with eight extended connective nodes were shown [Fig. 2(a)]. Although the same functions were provided as with the linear type, four function layers were displayed (microphone, speaker, projector, and digi-pen modes) at the same time, and the sublayered functions were hierarchically structured. Users could choose to operate specific functions by clicking on the related hierarchical function links and buttons [Fig. 2(b)]. Finally, for the network type of interface structure, four levels with 32 nodes were shown at one time [Fig. 3(a)]. Users could jump to any networked function as desired by clicking on any one of the 32 functions displayed directly [Fig. 3(b)].

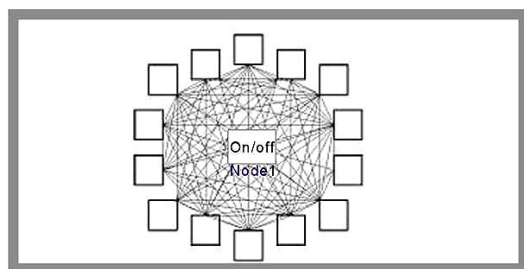


(b) Hierarchical topological structure

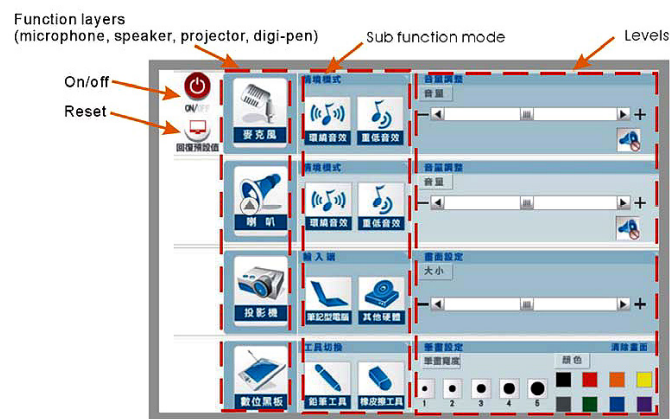


(a) Interface layout of hierarchical design

FIGURE 2 — The interface layouts of the hierarchical typological structure.



(b) Network topological structure



(a) Interface layout of network design

FIGURE 3 — The interface layouts of the network typological structure.

TABLE 1 — The summary of task specifications.

Task complexities	Task 1	Task 2	Task 3
Layers	1	2	3
Nodes	3	6	9

In total, three interactive tasks were planned, based on the number of layers and functional nodes of the topological structures. Stimuli were presented with three types of tasks, namely single-, double-, and triple-layered. Participants were asked to conduct all the interaction tasks pertinent to the digital-speech-desktop user-interface control.

The task specifications are illustrated in Table 1, and the features of the operation tasks are summarized below. First, a single-layered task with three necessary functional nodes was designed and labeled Task 1. The subjects were required to activate the requested functional mode and adjust the relative level of the specific function, such as volume, input device, or line breadth. Second, a double-layered task with six necessary functional nodes was designed and labeled Task 2. The subjects had to activate two of the requested functional nodes and adjust the relative levels of the specific functions. Finally, similar to Task 2 but with three requested multi-layer functions, a triple-layered task with nine necessary multi-functional nodes was designed and labeled Task 3. Subjects were required to conduct three of the requested functions and to adjust the relative level of a specific function to complete the task.

A 3 × 3 mixed-factor experimental design was adopted in this study. The independent variables were task type (single-, double-, and triple-layered) and topological structure (linear, hierarchical, and network). The dependent variable was the mean time taken to complete the tasks.¹³ A between-subjects design was adopted for the topological variables. Participants were randomly assigned to one of the experi-

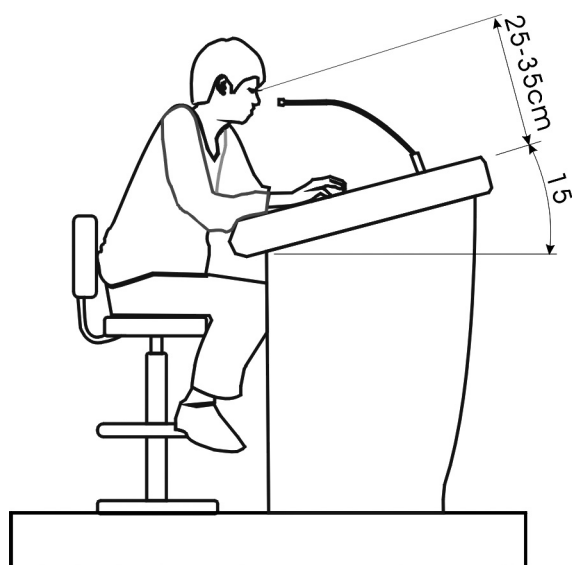


FIGURE 4 — The figure of situated participants and display.

mental conditions. A within-subjects design was used for the task type variables. All participants participated in all three tasks.

A multimedia room was used as the experiment site because it enabled reduced noise from the outside environment. A desktop PC and LCD monitor were used. The screen resolution was 1024×768 pixels, with a frame rate of 85 Hz. The screen size of the projection area was about 20×16 cm. Participants were asked to sit in front of the display at a distance of 25–35 cm (Fig. 4).

2.3 Procedure

Fourteen participants were randomly assigned to interact with each of the three topological structures (42 participants in total), so that each participant was required to interact with only one of the three types of user interface (linear, hierarchical, or network). The nature of the experiment and the task specifications were first briefly explained to each participant, and following this introduction, the participants were given 15 minutes to learn how to interact with the user interface. More time was given for participants who felt they needed further practice. The learning and practice session ended when participants had completed three randomly assigned practice tasks, one single-, one double-, and one triple-layered task. Instructions for these three tasks were given in writing on separate cards; participants took one card at a time and completed the task assigned on it before continuing with the task assigned by the next card; this continued until all three tasks were completed. Once the practice session ended, the experiment formally began after a 15-minute rest. Three tasks comprising the formal test were similarly given to participants on separate cards, and participants completed the task on one card before looking at the next. Each of the three tasks was thus taken randomly to avoid any carry-over effect. For both the practice and experimental trials, participants were instructed as follows: “Click on/off using the mouse to activate the interface, and the test will formally commence and be recorded. Your operation time is taken to reflect the effectiveness of the interface assigned to you, so please pay careful attention while performing the requested procedures. After reading

and memorizing the tasks, please try to complete the tasks as quickly as possible.” The full experiment lasted approximately 1 hour for each participant.

2.4 Data analysis

Task performance was analyzed in terms of speed (seconds) using a 3×3 mixed ANOVA analysis, which was conducted using SPSS software (SPSS, Inc., Chicago, IL, USA). The null hypothesis was rejected if the p value is equal to or less than 0.05.

3 Results

3.1 Overall results

The experimental data indicated that the main effect of search time for the within-subject factor, task type, was significant [$F(2, 78) = 122.78, p < 0.001$]; however, the main effect of the between-subject factor, topological structures, was not significant [$F(2, 39) = 0.211, p > 0.05$]. Finally, the two-way interactive effect was found to be significant [$F(4, 78) = 10.77, p < 0.001$].

3.2 Interactive effect

Figure 5 displays the mean task times for the three topological structures under each of the task-type conditions. The results of post hoc comparisons using Tukey’s HSD are shown in Table 2. The post hoc comparison revealed no significant difference in participants’ performance among the linear ($M = 18.85, SD = 2.39$), hierarchical ($M = 26.5, SD = 2.34$), and network ($M = 22.67, SD = 2.36$) types of topological structure on Task 1. On Task 2, participants’ task performance in the linear type ($M = 30.71, SD = 2.92$) was significantly better than that in the hierarchical ($M = 56.5, SD = 2.68$) and network types ($M = 48.45, SD = 6.37$) of topological structure. No significant difference was found between performance in the network and hierarchical types of topological structure on Task 2. In Task 3, participants’ task performance in the hierarchical ($M = 84.92, SD = 6.50$) and network ($M = 87.85, SD = 4.55$) types of topological structure was significantly better (less time consumed) than that in the linear type ($M = 128.07, SD = 13.45$) of structure. No significant difference was found between the hierarchical- and network-type structures in Task 3.

TABLE 2 — Post-hoc comparisons of interaction task time by means of Tukey’s HSD ($N = 42$).

Tasks	Topological structure	Search Time (sec)			Tukey’s HSD test
		<i>N</i>	<i>M</i>	Std. Deviation	
Task 1	Linear	14	18.85	2.39	
	Hierarchical	14	26.50	2.34	
	Network	14	22.67	2.36	
Task 2	Linear	14	30.71	2.92	A
	Hierarchical	14	56.50	2.68	B
	Network	14	48.45	6.37	B
Task 3	Linear	14	128.07	13.45	B
	Hierarchical	14	84.92	6.50	A
	Network	14	87.85	4.55	A

4 Discussion

4.1 Task complexity and user performance

The data generated from participants’ task performances revealed a significant difference in performance effectiveness on task type. The pairwise comparisons further showed the order of the search time in terms of the task type: Task 3 > Task 2 > Task 1. This may be because the task type

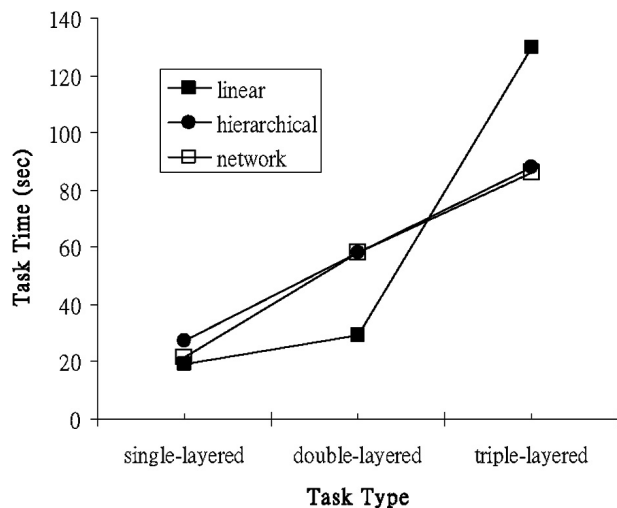


FIGURE 5 — Mean scores for task time of three topological structures for each type of task.

was planned based on the linked layers. As Task 1 was single layered, so simple that all participants could complete the task easily, thus less search time consumed; while task 2 was double-layered, and task 3 triple-layered; therefore, the search time of the tasks varied accordingly. Based on these findings, our research supports the first hypothesis, namely that task complexity would affect task performance among the interface users.

4.2 Interaction between task complexity and topological structure

Although the experiment data indicated that the main effect on topological structure was not significant, it was found that the interaction between task complexity and topological structure was significant. The post hoc comparison revealed that the effect of topological structure was not found on Task 1. On Task 2, participants performed best when interacting with a linear topological structure, as compared with the hierarchical or network topological structures. On Task 3, participants performed better with both network and hierarchical types of topological structures than with a linear structure. Tasks 2 and 3 were both multi-layered tasks, and yet produced opposite results; a linear topology resulted in better performance than network or hierarchical topologies on Task 2, while a contradictory result was found on Task 3. This may suggest that participants were able to more easily implement their complex mental model of Task 3 with the network and hierarchical topological structures than with the linear type of topological structure, but the opposite was true on Task 2 and there was no difference on Task 1. More specifically, when the task was single layered (Task 1) no difference was found among the three topological structures in terms of users' interface performance (search time), while a linear topological structure supported doubled-layered tasks (Task 2) better than did hierarchical or network topological structures; on the other hand, the hierarchical

and network topological structures facilitated triple-layered tasks (Task 3) better than did a linear topological structure.

Based on the discussion above, the second hypothesis, that there would be an interaction between topological structure and task complexity, was supported by the present study. This finding is also consistent with the argument that task characteristics have a significant interactive effect with different supportive interfaces.²² The linear structure sufficiently supports tasks that have little complexity, whereas the hierarchical and network structures best facilitate tasks that have greater complexity. Because no significant difference was found between the hierarchical and network topologies, these two structures can be further integrated as one category, namely, a hybrid topology. Thus, the next research step or application of design proposals or tests can focus on comparisons of linear and hybrid topological structures.

The present study extended the findings of previous studies on hypertext navigational performance.^{14,23–26} by demonstrating that different topological structures and task types interactively influence the performance of operations by interface users; *i.e.*, the existing phenomena and theories relating to hypertext or three-dimensional wayfinding have been shown to be valid in interface design as well. However, it is important to note that this research did not include potentially important variables such as gender and variations in wayfinding strategy^{27,28} that may also interactively affect the effectiveness and usability of an interface. In the future, an advanced experimental design is needed to explore the effects of related factors such as gender and wayfinding strategy.

5 Conclusion

This study adopted topological structures in the practice of digital-speech-desktop user-interface design. The research results indicate that, depending on the nature of the task complexity, each type of topology has unique design features that can facilitate users' performance. It was found that a linear topological structure can support tasks with less complexity well, whereas hierarchical and network topological structures better facilitate tasks with more complexity. These findings suggest that it may be important for two alternative or parallel control systems (linear and hybrid topologies) to be simultaneously available so that the user can shift between topological structures according to task complexity.

Acknowledgments

The authors thank the National Science Council of Taiwan for financially supporting this research under contract No. NSC 99-2410-H-130-042-.

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Wen-Te Chang received his doctoral degree in design from the Graduate School at National Taiwan University of Science and Technology in 2009. He has been a faculty member in the Department of Product Design at Ming Chuan University (Taiwan) since 1998. His research activities are in the areas of wayfinding, human-computer interaction design, and design research.



Chien-Hsiung Chen received his doctoral and master's degrees in industrial design (Ph.D. and MFA from the University of Kansas, Lawrence, Kansas, USA). He has been a faculty member in the Department of Industrial and Commercial Design at National Taiwan University of Science and Technology, Taipei, Taiwan. His cross-disciplinary research interests involve human factors in industrial/interaction design, color theories, and applications, design languages and representation, culture in industrial/interaction design, design methods and strategies, cognitive design, product/user interface usability engineering, visual interaction/communication design, design research methodology, human-computer interaction, and knowledge-based expert systems in design.